Context:

In the rapidly evolving digital financial landscape, the ability to detect fraudulent credit card transactions has become vital for minimizing financial losses and maintaining customer trust. With the significant increase in digital payment volumes, traditional rule-based detection systems often struggle to keep pace with increasingly sophisticated and adaptive fraud techniques employed by malicious actors. This inadequacy leads to the potential for substantial economic damage to financial institutions and a degradation of customer confidence. To address this pressing issue, financial institutions are increasingly turning to deep learning models that can analyze a variety of transaction features, including distance from the customer's home, spending patterns, and device usage behaviors. By leveraging these models, banks and financial service providers can facilitate real-time fraud detection, enhance their risk management strategies, and build a more resilient security framework. This proactive approach allows institutions to respond swiftly to potential fraud, effectively shielding both their assets and their customers from financial harm.

Objective:

The challenge at hand is to develop an effective deep learning model capable of accurately detecting fraudulent transactions amidst the complexities of transactional data. The objective is to create a system that can classify transactions as either genuine or fraudulent by utilizing various behavioral and transactional features to inform its predictions. The end goal of the project is to fortify the bank's defenses against fraud, fostering a secure and trustworthy environment for customers while preserving the institution's long-term viability and reputation in the competitive financial landscape.

Attribute Information:

- **distance_from_home**: Distance between transaction location and the customer's home.
- **distance_from_last_transaction**: Distance from the customer's last transaction.
- **ratio_to_median_purchase_price**: Ratio of the current transaction to the customer's median purchase price.
- **repeat_retailer**: Indicates if the retailer is one the customer has visited before.
- **used_chip**: Flag indicating whether the card chip was used.
- **used_pin_number**: Flag indicating whether a PIN was used.
- **online_order**: Flag indicating if the transaction was made online.
- **fraud**: Target variable; indicates if the transaction was fraudulent (1) or not (0).

Learning Outcomes:

- Exploratory Data Analysis
- Preparing the data to train a model
- Training a simple regression model using Neural Networks
- Making predictions using the model